

RegTrace-Agent: A Closed-Loop Framework for Evidence-Grounded Regulatory Response Review

Anonymous ACL submission

Abstract

Financial LLM systems are increasingly asked to decide whether an institution’s response actually resolves a prior review request, yet most financial benchmarks still evaluate static QA, extraction, or classification. We introduce RegTrace-Agent, a framework for evidence-grounded response resolution. Given a reviewer request, an organizational response, and revised artifacts, RegTrace-Agent retrieves evidence, decomposes the request into obligations, verifies visible gaps, constructs scalar/category/natural-language feedback, and optimizes a guarded reviewer policy. We instantiate the framework on SEC comment-letter review and construct RegTrace-SEC, 472 public EDGAR examples with amended-filing evidence, visible-evidence labels, feedback fields, and later same-topic SEC follow-up signals. Experiments show that amended evidence is necessary, full natural-language feedback yields the strongest reviewer among handwritten prompting, MIPROv2, and scalar/category GEPA variants, and a guarded verifier derived from the learned policy improves further. Later SEC follow-up comments provide external validation for the benchmark labels, suggesting that regulator-style feedback is a useful learning signal for evidence-grounded compliance reviewers.

1 Introduction

Financial NLP has rapidly moved toward LLM systems for long documents, domain-specific reasoning, and decision support. Models and benchmarks such as BloombergGPT, FinGPT, FinBen, FinanceBench, FinQA, TAT-QA, and ConvFinQA show both the promise of financial LLMs and the continuing difficulty of grounding answers in financial evidence (Wu et al., 2023; Yang et al., 2023; Xie et al., 2024; Islam et al., 2023; Chen et al., 2021; Zhu et al., 2021; Chen et al., 2022). Recent FinNLP work has similarly emphasized retrieval-augmented financial QA, SEC filing QA, financial

intelligence benchmarks, factuality judging, and evasive-answer detection (Lithgow-Serrano et al., 2025; Lai et al., 2025; Klimaszewski et al., 2025; Wu et al., 2025; Al Nuaimi et al., 2025). Most such settings, however, still present a static input and ask for a static answer, label, or report.

Many operational finance and compliance workflows have a different structure. A regulator, auditor, lender, or control function requests a correction, explanation, disclosure, or supporting artifact; the organization responds that it has revised an external document; the reviewer must decide whether visible evidence actually satisfies the request. We call this class of problems evidence-grounded response resolution. It is related to fact verification and document-level legal NLI (Thorne et al., 2018; Koreeda and Manning, 2021), but adds a prior reviewer request, a response that may only claim a revision, revised evidence that must be checked, and textual feedback about what remains missing.

Existing work leaves three limitations for this setting. First, benchmarks rarely ask whether an organization has repaired a documented deficiency. Second, response evaluation often has weak obligation grounding: a response can sound persuasive while amended evidence omits a named disclosure, quantification, exhibit, legal basis, or accounting analysis. Third, prompt and pipeline optimization often discards feedback structure by receiving only scalar success, although review failures are naturally described in text: what was requested, what was claimed, what evidence showed, and what gap remained.

We study these limitations through U.S. Securities and Exchange Commission (SEC) comment-letter review. SEC staff issue comments requesting expanded disclosure, clarification, revised accounting discussion, removal of unsupported statements, or additional exhibits; companies respond in writing, often with statements such as “the disclosure has been revised accordingly.” After review

completion, the SEC releases staff comment letters and company responses through EDGAR. These traces often point to amended filings and sometimes include later same-topic SEC follow-up comments. Accounting research treats comment letters as economically meaningful regulatory signals (Cassell et al., 2013; Bozanic et al., 2017; Baugh and Schmardebeck, 2023; Ryans, 2021); for NLP, they provide observable request-response-evidence-feedback traces.

We introduce RegTrace-Agent, a framework for evidence-grounded reviewers. It reconstructs review threads, retrieves revised-document evidence, decomposes requests into obligations, adjudicates visible evidence gaps, compiles scalar/category/full feedback, and optimizes a guarded reviewer policy. The learned reviewer is not a single hand-tuned prompt; it is the policy inside a system that gathers evidence, checks obligations, emits a guarded verdict, and uses adjudicated failures to improve the next offline reviewer.

This design makes response resolution a strong testbed for language-feedback optimization. Prompt-programming work treats LLM pipelines as declarative programs whose prompts and demonstrations can be optimized (Khattab et al., 2023); GEPA argues that natural-language feedback can expose reusable failure structure that scalar reward discards (Agrawal et al., 2026). In our setting, that structure is regulator-style: a mistake can be explained as a missing requested disclosure, unsupported revision claim, absent exhibit, or incomplete analysis.

The above limitations motivate three research questions. RQ1 asks whether public review traces can be converted into a benchmark for evidence-grounded response resolution. RQ2 asks whether natural-language feedback improves reviewer optimization beyond scalar prompt search and coarse categories. RQ3 asks whether the resulting decisions generalize across threads, time, and issue categories, and whether unresolved decisions are corroborated by later regulator follow-up.

We make four contributions. First, we define evidence-grounded response resolution and introduce RegTrace-Agent, a feedback-learning framework for evidence retrieval, obligation checking, feedback construction, and reviewer optimization. Second, we instantiate it as RegTrace-SEC, a public EDGAR benchmark with SEC comments, company responses, amended-filing evidence, labels, and feedback fields. Third, we show that GEPA

with full natural-language feedback outperforms handwritten prompting, MIPROv2, scalar/category GEPA variants, and a supervised lexical baseline; a guarded verifier distilled from the learned policy improves further. Fourth, we provide evidence ablations, independent-feedback and feedback-component ablations, prompt-evolution analysis, and systematic corroboration from real next-round SEC follow-up comments.

2 Related Work

Financial NLP and evidence-grounded evaluation. Financial NLP has moved from broad document classification toward long-document reasoning, evidence grounding, and compliance-sensitive decisions. FinQA, TAT-QA, and ConvFinQA introduced financial reasoning questions over reports, tables, and conversations (Chen et al., 2021; Zhu et al., 2021; Chen et al., 2022); FinanceBench evaluates open-book QA over company documents (Islam et al., 2023). Recent FinNLP work has similarly emphasized RAG over financial documents, accessible finance-intelligence benchmarks, factuality and causality judging, SEC filing QA, and evasive-answer detection (Lithgow-Serrano et al., 2025; Klimaszewski et al., 2025; Lai et al., 2025; Al Nuaimi et al., 2025; Wu et al., 2025). Our task differs by asking whether retrieved evidence satisfies a prior regulator request.

SEC comment letters and regulatory review. Accounting and disclosure research has studied SEC comment letters as regulatory signals, including remediation costs, disclosure effects, reducing disclosure deficiencies, and textual classification of comment-letter topics (Cassell et al., 2013; Bozanic et al., 2017; Baugh and Schmardebeck, 2023; Ryans, 2021). That literature generally treats comment letters as economic or disclosure events. We instead treat the correspondence as document-grounded supervision for LLM systems. Each review unit contains a staff request, company response, amended filing evidence, and sometimes a next-round SEC follow-up.

Prompt optimization and language feedback. DSPy reframes LLM pipelines as declarative programs whose prompts and demonstrations can be optimized against metrics (Khattab et al., 2023). MIPROv2 and related optimizers primarily use scalar feedback. A broader line studies language feedback, including self-refinement, verbal reinforcement,

186 enforcement, and critique-based training (Scheurer
187 et al., 2022; Madaan et al., 2023; Shinn et al., 2023).
188 GEPA uses natural-language reflection over execu-
189 tion traces (Agrawal et al., 2026). This distinction
190 is central here because SEC response errors are
191 naturally textual: missing disclosures, unsupported
192 revision claims, absent exhibits, incomplete analy-
193 sis, or over-demand beyond the SEC request.

194 3 RegTrace-Agent Framework

195 3.1 General Review-Trace Setting

196 RegTrace-Agent applies to workflows where a re-
197 sponse claims to satisfy a prior review request by
198 modifying or supplying evidence outside the re-
199 sponse itself. Formally, each review trace contains
200 a request q , a response r , a collection of revised
201 or supporting artifacts $\mathcal{D}$, and optional later re-
202 viewer feedback u^+ . The system retrieves evidence
203 $e = R(q, r, \mathcal{D})$, extracts obligations $o_{1:m} = O(q)$,
204 and applies a verifier $V(q, r, e, o_{1:m})$ whose output
205 is either resolved or unresolved. The same trace
206 can also produce feedback for learning. A scalar
207 signal says whether the prediction is correct. A
208 category signal names a broad issue or gap type. A
209 full natural-language signal describes the request,
210 the response action, the visible evidence, and any
211 unmet requirement. This makes the framework
212 applicable beyond SEC review whenever request-
213 response-evidence traces are available.

214 3.2 SEC Instantiation

215 Each SEC example contains three test-time
216 fields: `sec_comment`, `company_response`, and
217 `retrieved_snippets` from amended filings. The
218 output is binary. `resolved` means visible amended
219 evidence covers the material request; `unresolved`
220 means a named item, quantification, exhibit, docu-
221 ment, legal analysis, accounting analysis, or other
222 material requirement remains missing. The label is
223 a visible-evidence judgment, not a legal determina-
224 tion or an inference from absence of later follow-
225 up.

226 3.3 SEC Trace Construction

227 Figure 1 separates the runtime reviewer from the
228 feedback-learning loop. In the SEC instantiation,
229 we harvest UPLOAD and CORRESP materials from
230 EDGAR, group staff comments and company re-
231 sponses into review threads, and select cases where
232 the company claims a disclosure was added, re-
233 vised, removed, or filed elsewhere. We identify

Quantity	Count
Examples	472
SEC review threads	109
Companies	95
Resolved / unresolved	153 / 319
Examples with verified follow-up	236
Issue categories	7
Visible-evidence gap types	8

Table 1: Benchmark scale. Issue categories include other disclosure issues, risk factors, SPAC disclosure, revenue recognition, crypto, non-GAAP, and cybersecurity.

amended filing candidates using filing metadata, 234
amendment references, page and section cues, and 235
response text. Candidate filings are drawn from 236
a window of 10 days before to 60 days after the 237
response date, with at most 8 candidate filings per 238
row. A local lexical ranker retrieves top chunks, 239
and an LLM reranker/adjudicator evaluates the top 240
3 snippets. From 589 revision-claim cases, 588 241
have candidate filings and snippets; 472 remain 242
after requiring visible evidence that directly or par- 243
tially addresses the SEC request. We then adjudi- 244
cate whether visible evidence satisfies the request 245
and construct labels plus feedback fields. 246

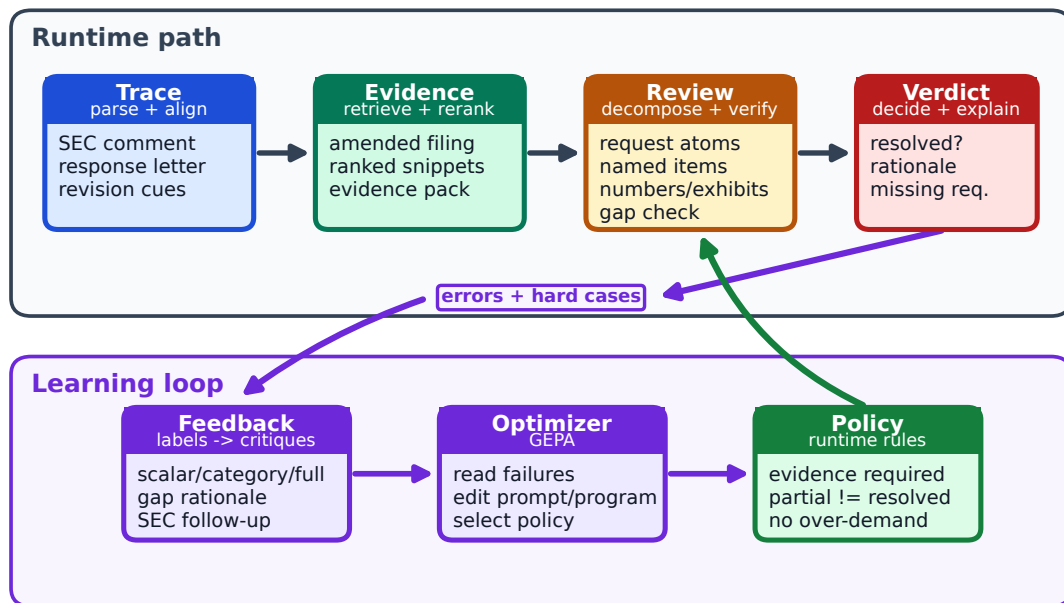
The retrieval component gives the agent evi- 247
dence to inspect. The adjudication and feedback 248
layer gives it something to learn from: what the 249
reviewer asked for, what the company claimed to 250
do, what the visible evidence supports, and what 251
remains missing. These adjudication fields are not 252
shown as ordinary test-time inputs. They are used 253
for optimization feedback, audits, and analysis. 254

255 3.4 Scale, Categories, and Splits

RegTrace-SEC contains 472 examples from 109 256
SEC review threads and 95 companies. The bench- 257
mark focuses on difficult revision-claim cases, and 258
the label distribution is operationally realistic: 153 259
examples are resolved, and 319 are unresolved. 260
Table 1 summarizes the corpus and split regimes. 261

The corpus covers seven issue categories and 262
eight visible-evidence gap types; Appendix B gives 263
the full distribution. We use grouped, temporal, and 264
topic-holdout regimes. The grouped split holds out 265
entire review threads (229/104/139 train/dev/test). 266
The temporal split trains on 2022–2023 and tests 267
on 2024–2025 (264/64/144). The topic-holdout 268
split tests transfer to held-out issue categories 269
(279/68/125), including crypto and non-GAAP 270
cases absent from training categories. 271

RegTrace-Agent: closed-loop evidence review and feedback optimization



Test-time inputs: SEC comment, company response, retrieved amended evidence. Feedback fields are train/audit only.

Figure 1: RegTrace-Agent architecture. The runtime path parses a review trace, retrieves amended-filing evidence, checks request obligations, and emits a guarded verdict. The learning loop stores scalar, category, and natural-language feedback, lets GEPA revise the reviewer policy, and reinjects the learned policy into the obligation-review step. At test time the reviewer sees only the SEC comment, company response, and retrieved amended evidence.

3.5 From Labels to Feedback

A standard benchmark would stop at binary labels. Our construction preserves the textual structure of each decision. For every example, adjudication records the material SEC request, the company action, evidence relevance, supporting evidence or resolution basis, visible unmet requirement, and a short rationale. This lets us compare increasingly rich optimization signals:

- scalar: correct or incorrect;
- category: correct/incorrect plus coarse issue or gap type;
- full: natural-language feedback describing the SEC request, company action, evidence coverage, and unmet requirement.

Full feedback can state that the evidence discusses combined revenue projections but fails to disclose entity-level operating costs, product pricing, gross margins, and forecast limitations. This is the reusable decision rule that scalar metrics discard.

4 Methods

4.1 Prediction Setup

The prediction model receives the SEC comment, company response, and raw retrieved amended-filing snippets. It returns a binary label and a short rationale. Explicit resolved is scored as resolved, unresolved, partially_resolved, and other insufficient labels are normalized to unresolved. The rationale is required for inspection but not directly rewarded.

4.2 Optimization Ladder

We compare five prompt methods. Baseline prompt is a handwritten instruction. MIPROv2 is scalar prompt optimization. GEPA-scalar uses reflective prompt optimization with correctness-only feedback. GEPA-category adds coarse category feedback. GEPA-full uses full natural-language adjudication feedback. We also include a TF-IDF logistic regression baseline and an independent-feedback GEPA ablation for reviewer-facing robustness.

All methods receive the same evidence at test time, isolating the optimization signal rather than evidence access. The key comparison is whether natural-language feedback adds value beyond

prompt search itself.

4.3 Implementation Details

All main prompt-optimization runs use openai/gpt-4o-mini as the frozen prediction model with temperature 0 and 500 output tokens. GEPA uses the same model as the reflection model with temperature 0.7 and 1500 reflection tokens, seed 17, max_metric_calls=150, reflection minibatch size 4, Pareto candidate selection, and 4 evaluation threads. MIPROv2 uses the same prediction and prompt models, 8 trials, 4 prompt candidates, and at most 4 labeled and 4 bootstrapped demonstrations. The evidence reranker/adjudicator also uses gpt-4o-mini at temperature 0. Model aliases rather than dated provider snapshots are recorded in the released artifacts.

4.4 Optimization as Feedback-Conditioned Search

We view each method as search over a prompt or prompt program p . Given an input x and label y , the frozen LLM defines $\hat{y}_p(x) = f_\theta(x; p)$, and optimization seeks a prompt with high validation reward. The methods differ in what they observe after an error. Scalar optimization observes only correctness. Category feedback adds a coarse gap type. Full feedback adds a short critique describing the request, company action, visible evidence, and unmet requirement.

The intuition is information-theoretic. Let C be the latent cause of failure, such as missing quantification or unsupported revision claim. Full feedback should help when it lowers uncertainty about C , because the optimizer can propose a more targeted semantic edit. This is not a general guarantee: a noisy or misaligned language channel can hurt. The hypothesis of RegTrace-Agent is that review traces are unusually well suited to language feedback because the failure causes are naturally written as reviewer critiques.

4.5 What the Optimizer Sees

During optimization, each candidate prompt is evaluated on development examples. When a prediction is incorrect, the metric returns feedback in one of three forms:

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scalar: incorrect
category: incorrect;
gap_type=missing_specific_disclosure
full: incorrect. The SEC requested
X. The company claimed Y. The visible

```

Input	Acc.	Ma-F1	Res-F1	Unres-F1
Response only	0.712	0.575	0.333	0.817
+ raw snippets	0.784	0.748	0.651	0.844
+ oracle summary	0.964	0.956	0.938	0.975

Table 2: Evidence ablation. Adding raw amended-filing snippets raises macro-F1 by 17.3 points. The oracle summary is an upper bound, not a deployable baseline.

evidence shows Z, but does not show
required item W.

Full feedback summarizes the adjudicated relationship between the SEC request, response, and retrieved evidence for optimization examples only. It does not reveal hidden evidence at test time; any advantage must come from a better learned checklist for visible-evidence review.

5 Experiments

5.1 Evidence Ablation

We first test whether amended-filing evidence matters. Table 2 evaluates a response-only condition, a raw-evidence condition, and an oracle evidence summary condition on the grouped held-out test set.

Response-only prompting cannot verify whether a revision claim is supported. Raw snippets nearly double resolved F1, while the oracle summary shows that much of the remaining difficulty is interpreting snippets against a specific request.

5.2 Main Prompt-Optimization Results

Table 3 reports the final prompt-optimization ladder on the frozen grouped split. All methods receive the same test-time context.

The majority baseline has high accuracy because unresolved examples dominate, but zero resolved F1. The handwritten prompt is conservative: it has perfect unresolved precision but misses most gaps. MIPROv2 improves substantially, GEPA-scalar improves recall, and GEPA-category is too coarse in this run. GEPA-full is strongest overall, improving macro-F1 by 10.4 points over MIPROv2 and 8.1 points over GEPA-scalar. A TF-IDF logistic regression classifier trained on the same raw evidence reaches only 0.523 macro-F1, while an oracle-summary variant reaches 0.858, indicating that lexical supervision does not explain the GEPA full gain and that evidence interpretation is the bottleneck.

We also compute five-fold out-of-fold results over review-thread groups. Mean macro-F1 is

Method	Acc.	Macro-F1	Res-F1	Unres-F1	Unres-P	Unres-R
Majority unresolved	0.691	0.409	0.000	0.817	0.691	1.000
Baseline prompt	0.424	0.402	0.518	0.286	1.000	0.167
MIPROv2	0.655	0.652	0.619	0.684	0.929	0.542
GEPA-scalar	0.705	0.675	0.577	0.773	0.824	0.729
GEPA-category	0.597	0.595	0.562	0.627	0.870	0.490
GEPA-full	0.784	0.756	0.674	0.839	0.867	0.812

Table 3: Frozen grouped-split leaderboard. GEPA-full gives the best macro-F1 and unresolved F1 under the same raw-evidence test-time context.

Reviewer	Macro-F1	Res-F1	Unres-F1
Monolithic reviewer	0.645	0.614	0.676
Obligation-guided reviewer	0.619	0.613	0.624
GEPA-full optimized prompt	0.756	0.674	0.839
RegTrace guarded verifier	0.801	0.727	0.874

Table 4: Reviewer-structure ablation. The guarded verifier is a policy-distillation prototype derived from the obligation-evidence policy exposed by GEPA-full prompt evolution.

0.698 for GEPA-full, 0.647 for MIPROv2, 0.609 for GEPA-scalar, 0.545 for GEPA-category, and 0.446 for the handwritten baseline. The fold standard deviations are nontrivial (0.151 for GEPA-full and 0.121 for MIPROv2), so we treat the single grouped split as a headline comparison and the out-of-fold analysis as stability evidence rather than as a deterministic margin claim. To test possible feedback leakage, we regenerate full textual feedback with an independent gpt-5.4-mini writer that sees only test-time fields plus the gold label, not the original adjudication rationales; GEPA with this feedback reaches 0.750 macro-F1, nearly matching the original 0.756.

5.3 Reviewer-Structure Ablation

The optimization ladder tests feedback richness. We also ask whether reviewer structure matters. Table 4 compares a monolithic judgment, an obligation-guided judgment, and a guarded verifier that requires every material obligation to be visibly satisfied, with GEPA-full as the optimized-prompt reference.

Simply listing obligations is not enough; the gain comes from the guard that requires visible support for each material request. We report this as a policy-distillation prototype rather than the primary optimizer comparison, since it adds human-designed structure after prompt-evolution analysis.

5.4 Feedback Component Ablation

Table 5 holds test-time input fixed and changes only optimization-time feedback.

Request-action-only feedback induces aggressive skepticism and the highest unresolved recall, but sacrifices resolved F1. Full feedback teaches both sides of the boundary: unmet requirements for unresolved cases and supporting evidence for resolved cases.

5.5 Temporal and Topic Generalization

Table 6 shows that the optimized prompt transfers beyond the grouped split. In the temporal setting, GEPA-full trains/tunes on 2022–2023 and tests on 2024–2025 material. In the topic setting, test examples include crypto and non-GAAP cases absent from training issue categories.

The topic-holdout result argues against a curated topic explanation. Per-topic breakdowns show GEPA-full improves over MIPROv2 on crypto, non-GAAP, revenue recognition, risk factors, and broad other-disclosure cases, with largest gains in revenue recognition and non-GAAP.

6 External SEC Follow-Up Corroboration

Visible-evidence labels come from an adjudication pipeline, so we test whether they align with later regulator behavior. Verified same-topic next-round SEC follow-up is not the primary label, but it is a useful corroboration signal because it can restate or narrow an unresolved gap.

We additionally audit 60 examples with an independent LLM auditor that receives the SEC comment, company response, and visible amended evidence but not the original label rationale. The audit agrees with the benchmark label in 49 of 60 cases, with mean confidence 0.897. We report this as an audit of visible-evidence label consistency, not as expert legal/accounting validation.

Among 236 examples with verified follow-up, unresolved cases are corroborated at 68.1%, while

Feedback	Acc.	Macro-F1	Res-F1	Unres-F1	Unres-P	Unres-R
Full feedback	0.784	0.756	0.674	0.839	0.867	0.812
No evidence rationale	0.712	0.682	0.583	0.780	0.826	0.740
No unmet requirement	0.741	0.708	0.609	0.806	0.833	0.781
Request + action only	0.784	0.724	0.595	0.853	0.806	0.906

Table 5: Feedback component ablation. Request-action feedback induces aggressive unresolved detection, while full feedback gives the best macro-F1 by calibrating skepticism with evidence rationales.

Split	Method	Acc.	Macro-F1	Unres-F1	Unres-R
Time	Baseline	0.451	0.434	0.336	0.206
	MIPROv2	0.521	0.519	0.489	0.340
	GEPA-scalar	0.653	0.650	0.679	0.546
	GEPA-full	0.826	0.791	0.877	0.918
Topic	Baseline	0.440	0.440	0.426	0.274
	MIPROv2	0.576	0.565	0.635	0.484
	GEPA-scalar	0.528	0.524	0.569	0.411
	GEPA-full	0.784	0.734	0.849	0.800

Table 6: Temporal and topic-holdout generalization. GEPA-full transfers substantially better than scalar optimization.

Group	N	D/P	Rate
Unres. + f/u	160	109	68.1%
Res. + f/u	76	7	9.2%

Table 7: Real follow-up corroboration. Later same-topic SEC comments directly or partially corroborate unresolved visible-evidence labels far more often than resolved labels.

Method	Claim insuff.	Evidence required	Request match	Missing items
GEPA-full	2/2	2/2	2/2	2/2
GEPA-scalar	2/2	2/2	0/2	0/2
GEPA-category	2/2	2/2	0/2	0/2
Req.+action only	2/2	2/2	2/2	2/2

Table 8: Rule presence in optimized prompt candidates. Full feedback more consistently induces named-request matching and missing-item reasoning.

resolved cases are corroborated at only 9.2% (Table 7). Follow-up availability itself is balanced across labels (49.7% for resolved and 50.2% for unresolved; Fisher exact $p = 1.0$), so this is not merely an availability artifact.

Following reviewer-style paired analysis, we also treat later follow-up as a noisy external reference on a clean 220-example subset. The benchmark label reaches 0.756 macro-F1 against this signal, compared with 0.590 for five-fold out-of-fold GEPA-full predictions and 0.460 for the handwritten baseline. We therefore use follow-up primarily as label-validity evidence, not as proof that every GEPA-full win is separately corroborated beyond the unresolved base rate.

In one expense-disclosure case, the SEC asked for material components and trend discussion. The company claimed a revision, but retrieved evidence showed only a line item and totals; GEPA-full predicted unresolved, and the next-round SEC follow-up again requested the component breakout. SPAC transaction-background cases show the same pattern for missing negotiation details.

7 Prompt Evolution and Case Studies

7.1 Prompt Evolution

Prompt evolution shows that GEPA-full does not merely make the prompt longer. It turns adjudication feedback into a compact regulatory reading policy. The seed prompt is a single sentence; the final GEPA-full run saves optimized candidates of 349 and 397 words (Appendix A).

Scalar and category feedback learn general caution: do not trust a revision claim unless evidence supports it. Full feedback more consistently learns named-request matching and missing-item reasoning (Table 8), shifting from generic skepticism to obligation-level comparison: SEC request $\rightarrow$ company action $\rightarrow$ visible evidence $\rightarrow$ remaining gap.

7.2 Method-Disagreement Cases

The strongest GEPA-full wins involve concrete request-evidence mismatches. In a financial-projections case, the SEC requested individual company forecast information and material assumptions; the amended evidence showed only combined revenue projections. The baseline treated

519 this as sufficient, while GEPA-full marked it unre-
520 solved because the requested assumption set was
521 still absent.

522 In a crypto terminology case, the amended evi-
523 dence defined stablecoins but did not explain the
524 SEC’s requested contrast with unstable cryptocur-
525 rency. Baseline, MIPROv2, GEPA-scalar, and
526 GEPA-category predicted resolved; GEPA-full
527 predicted unresolved. In a PIPE financing case,
528 MIPROv2 treated the presence of price terms as
529 sufficient; GEPA-full required equivalent price-per-
530 share disclosure for each agreement, a gap later
531 echoed in SEC follow-up.

532 GEPA-full also has informative errors. In one
533 ownership-percentage case, retrieved evidence con-
534 tained calculations under redemption scenarios and
535 the benchmark label was resolved. GEPA-full
536 predicted unresolved because it wanted a more
537 explicit cover-page presentation. This reveals the
538 main calibration risk: full-feedback prompts learn
539 to be skeptical of unsupported revision claims, but
540 can over-require exact presentation when visible
541 evidence substantially satisfies the request. Ap-
542 pendix C lists additional analyzed cases.

543 8 Discussion

544 The main result is not simply that one optimizer
545 beats another. RegTrace-Agent turns a recur-
546 ring compliance pattern into an offline feedback-
547 learning system: collect the trace, retrieve the re-
548 vised artifact, check obligations against visible ev-
549 idence, issue a guarded verdict, and recycle ad-
550 judicated failures as feedback. The output is not
551 only a binary label, but a traceable unresolved-gap
552 rationale.

553 Language feedback helps because it preserves
554 decision structure. A scalar signal says only that
555 a verdict failed; full feedback says which named
556 request, company action, amended snippet, and
557 remaining obligation were mismatched. The op-
558 timized prompt is consequently an interpretable
559 policy artifact: extract material requests, distrust
560 unsupported revision claims, inspect evidence for
561 named items, avoid over-requiring beyond the reg-
562 ulator’s request, and distinguish partial disclosure
563 from resolution.

564 SEC correspondence is the public setting where
565 we can observe request, response, revised filing,
566 and later reviewer behavior, but the trace structure
567 is broader. Audit remediation, regulatory reporting,
568 lending-document stipulations, data-quality excep-

tion handling, and internal compliance monitoring 569
also contain request-response-evidence loops. The 570
transferable claim is that such traces can support 571
retrieval, obligation checking, adjudication feed-572
back, optimized reviewing, and hard-case mining; 573
the limitation is that high-recall language-feedback 574
policies can become over-skeptical. 575

576 9 Conclusion

We introduced RegTrace-Agent, a framework for 577
evidence-grounded regulatory response review, and 578
instantiated it as RegTrace-SEC using public SEC 579
comment-letter traces. Across grouped, tempo-580
ral, and topic-holdout evaluations, full natural-581
language feedback yields the strongest optimized 582
reviewer. Evidence ablations, prompt evolution, 583
and real SEC follow-up corroboration suggest that 584
financial NLP benchmarks can move from static 585
labels toward feedback-rich review systems that 586
preserve the reasons behind expert decisions. 587

588 Limitations and Data Statement

This study is conditioned on visible retrieved ev-589
idence, so missed snippets can affect labels. The 590
benchmark is LLM-adjudicated, with independent 591
audit and external follow-up corroboration but with-592
out a full expert legal/accounting audit. Same-topic 593
SEC follow-up is a noisy behavioral signal, not 594
ground truth. We frame labels as visible-evidence 595
judgments rather than legal determinations. The 596
release includes benchmark JSONL, splits, acces-597
sion identifiers, retrieval metadata, snippets, labels, 598
feedback fields, prompt versions, outputs, and anal-599
ysis scripts. 600

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737	A Full Optimized Prompts	
738	The main paper quotes short excerpts from opti-	
739	mized prompts. Below are the full GEPA-full op-	
740	timized candidates exposed by the final grouped-	
741	split run. The seed prompt was: “Classify whether	
742	a company resolved an SEC comment using the	
743	visible evidence.”	
744	A.1 GEPA-full Optimized Candidate 1	
745	You are tasked with classifying whether a com-	
746	pany has resolved a comment from the SEC based	
747	on the provided evidence. Follow these instruc-	
748	tions carefully:	
749	1. Input Format: You will receive three types of	
750	inputs: <code>sec_comment</code> , <code>company_response</code> , and	
751	<code>amended_evidence</code> .	
752	2. Classification Criteria: Determine if the com-	
753	pany has fully resolved the SEC comment by ana-	
754	lyzing the company response in conjunction with	
755	the amended evidence. If the response satisfies	
756	the SEC’s request and the evidence supports this	
757	claim, classify the comment as resolved. If the re-	
758	sponse does not meet the SEC’s request or if there	
759	are still outstanding issues, classify the comment	
760	as unresolved.	
761	3. Evaluation Process: Carefully compare the	
762	SEC request against the company’s response and	
763	the amended evidence. Look for specific items or	
764	information requested by the SEC, quantification	
765	or detailed analysis that meets SEC requirements,	
766	references to regulatory guidance or previous de-	
767	cisions, and clear explanations of how the com-	
768	pany adjusted disclosures or practices. Take note	
769	of visible unmet requirements, partial fixes, or	
770	lack of supporting documentation in the amended	
771	evidence.	
	4. Output Format: Include a reasoning section,	772
	a label indicating resolved or unresolved, and a	773
	concise reason.	774
	5. Feedback Incorporation: Be vigilant about	775
	claims made by the company; ensure that	776
	amended evidence corroborates these claims.	777
	Identify instances where the response may par-	778
	tially address SEC concerns but still falls short of	779
	complete compliance.	780
	A.2 GEPA-full Optimized Candidate 2	781
	You are tasked with classifying whether a com-	782
	pany has resolved a comment issued by the SEC	783
	based on provided evidence. Follow these specific	784
	instructions for the task:	785
	1. Input Format: You will receive <code>sec_comment</code> ,	786
	<code>company_response</code> , and <code>amended_evidence</code> .	787
	The SEC comment is the original request for clar-	788
	ification or additional information. The company	789
	response indicates what changes were made or	790
	explanations provided. The amended evidence	791
	contains snippets from the amended filing or other	792
	documents that support the response.	793
	2. Classification Criteria: Determine whether the	794
	company’s response and amended evidence ade-	795
	quately address the SEC’s concerns as outlined	796
	in the original comment. A response is resolved	797
	if it fully satisfies the requirements of the SEC	798
	comment, including clear explanations, specific	799
	details, and quantifications when necessary. A	800
	response is unresolved if there are still gaps or	801
	missing elements in the company’s explanation or	802
	if amended evidence does not address all parts of	803
	the SEC comment.	804
	3. Evaluation Process: Carefully analyze the SEC	805
	comment and identify the specific details or expla-	806
	nations requested. Review the company response	807
	to see if the company acknowledged the comment	808
	and provided a meaningful response. Examine	809
	amended-evidence snippets to confirm that they	810
	support the company’s claims and cover all re-	811
	quested aspects of the SEC comment. Look for	812
	direct references in the amended evidence that	813
	address the SEC’s specific requests, such as own-	814
	ership structures, accounting adjustments, or en-	815
	forcement risks.	816
	4. Labeling: If the company has fully addressed	817
	the SEC’s concerns with adequate evidence, label	818
	the response resolved. If there are omissions or	819
	inadequacies in addressing SEC requests, label it	820
	unresolved.	821
	5. Justification: Provide clear and concise rea-	822
	soning that cites specific elements from the SEC	823
	comment, company response, and amended evi-	824
	dence. Highlight specific unmet requirements or	825
	gaps in the evidence that contribute to an unre-	826
	solved classification.	827
	6. Feedback Incorporation: Pay attention to feed-	828
	back from previous examples to identify common	829
	pitfalls, such as failing to compare the SEC re-	830
	quest against the evidence adequately or missing	831
	key elements in the response.	832
	B Additional Dataset Details	833
	The most common issue categories are other disclo-	834
	sure issues (183), risk factors (91), SPAC disclosure	835

836 (78), revenue recognition (75), crypto (28), non-
837 GAAP (16), and cybersecurity (1). The most com-
838 mon gap types are resolved visible evidence (153),
839 missing specific disclosure (140), missing quantifi-
840 cation (59), resolved-but-regulator-requested-more-
841 detail (38), missing exhibit or document (29), in-
842 complete accounting analysis (25), incomplete le-
843 gal or regulatory analysis (25), and other visible
844 evidence gap (3).

845 **C Additional Qualitative Cases**

846 The qualitative appendix used during analysis in-
847 cludes full SEC comments, company responses,
848 amended evidence snippets, method predictions,
849 GEPA-full rationales, and later SEC follow-up text
850 for selected cases. These include stablecoin termi-
851 nology, financial forecast assumptions, PIPE price-
852 per-share disclosure, Item 404 related-party disclo-
853 sure, expense breakout and trend discussion, and
854 SPAC transaction-background disclosure.